

INFORMATION TECHNOLOGY INSTITUTE

LAB 7

5T - OTA





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Part 1: OTA Design

Sizing

Input pair sizing

Parameter	Value
Supply voltage	1.8V
Load	5pF
Open loop DC voltage gain	$\geq 34\text{dB}$
CMRR @ DC	$\geq 74\text{dB}$
Phase margin (PM)	$\geq 70^\circ$
Current seed (reference current)	10uA
CM input range – low	$\leq 1.2\text{V}$
CM input range – high	$\geq 1.5\text{V}$
GBW	$\geq 5\text{MHz}$
OTA current consumption	Minimize
Area	Minimize

$$GBW = \frac{gm_{1,2}}{2\pi CL}$$

$$gm_{1,2} = 157 \mu S$$

For optimal power the used current will be $I_D = 10 \mu A$, $\frac{gm_{1,2}}{I_D} = 15.7$

$$A_{ol} = 50.12 = gm_{1,2} * (r_{o2} || r_{o4})$$

Because the active load have a lower $\frac{gm}{I_D}$ we wont assume that $r_{o2} = r_{o4}$

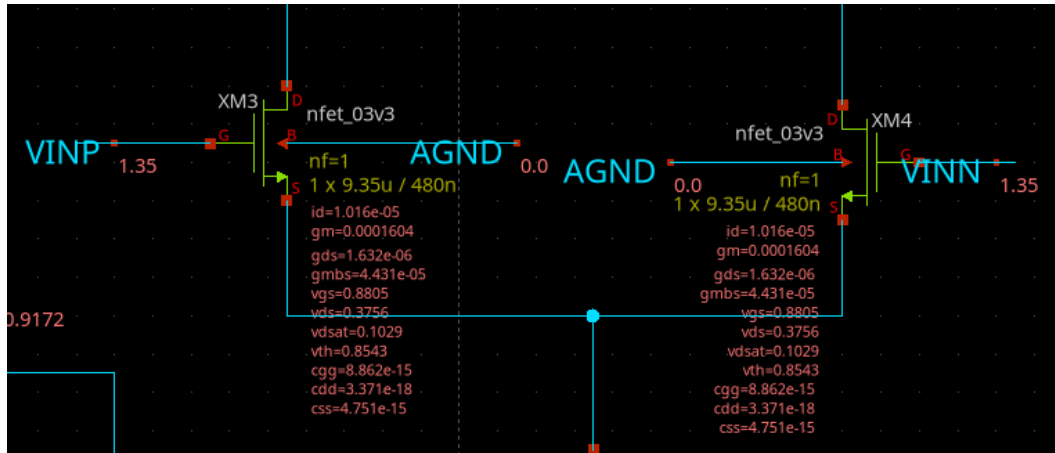
And r_{o4} will be bigger than r_{o2} so we will assume r_{o4} larger than r_{o2} by 10%

We need a total Rout 625k to achieve the $A_{ol} = 50.12$ so let $r_{o4} = 1.1 * 625k = 690k$

So we $r_{o2} = 590k$



and because of the decreased V_{DS} and increased V_{th}
I needed to increase the width a bit to achieve the required
Gain so the width after fine tuning became 9.3 μm



Plot 3A Plot 3B Plot 3C Plot 3D

Import: Plot 1A OK ?

LUT Settings

LUT nmos_03v3 ?

Corner TT ? All ?

Temp (°C) 27.0 ? All ?

Frequency 1 ?

ID 10u ?

gm/ID 15.7 ?

ro 590k ?

VDS 0.37 ?

VSB 0.4 ?

Stack 1 ?

Results:

	Name	TT-27.0
1	ID	10u
2	IG	N/A
3	L	480n
4	W	8.56u
5	VDB	770m
6	VGB	1.266
7	VGS	866.5m
8	VDS	375.6m

Y-Expr gm/ID*IT ?

Plot ?

Verify ?

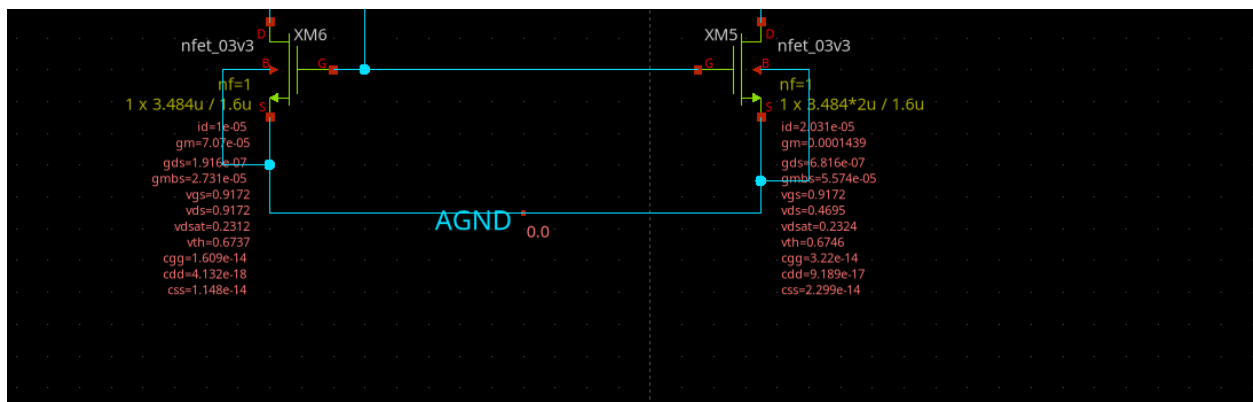
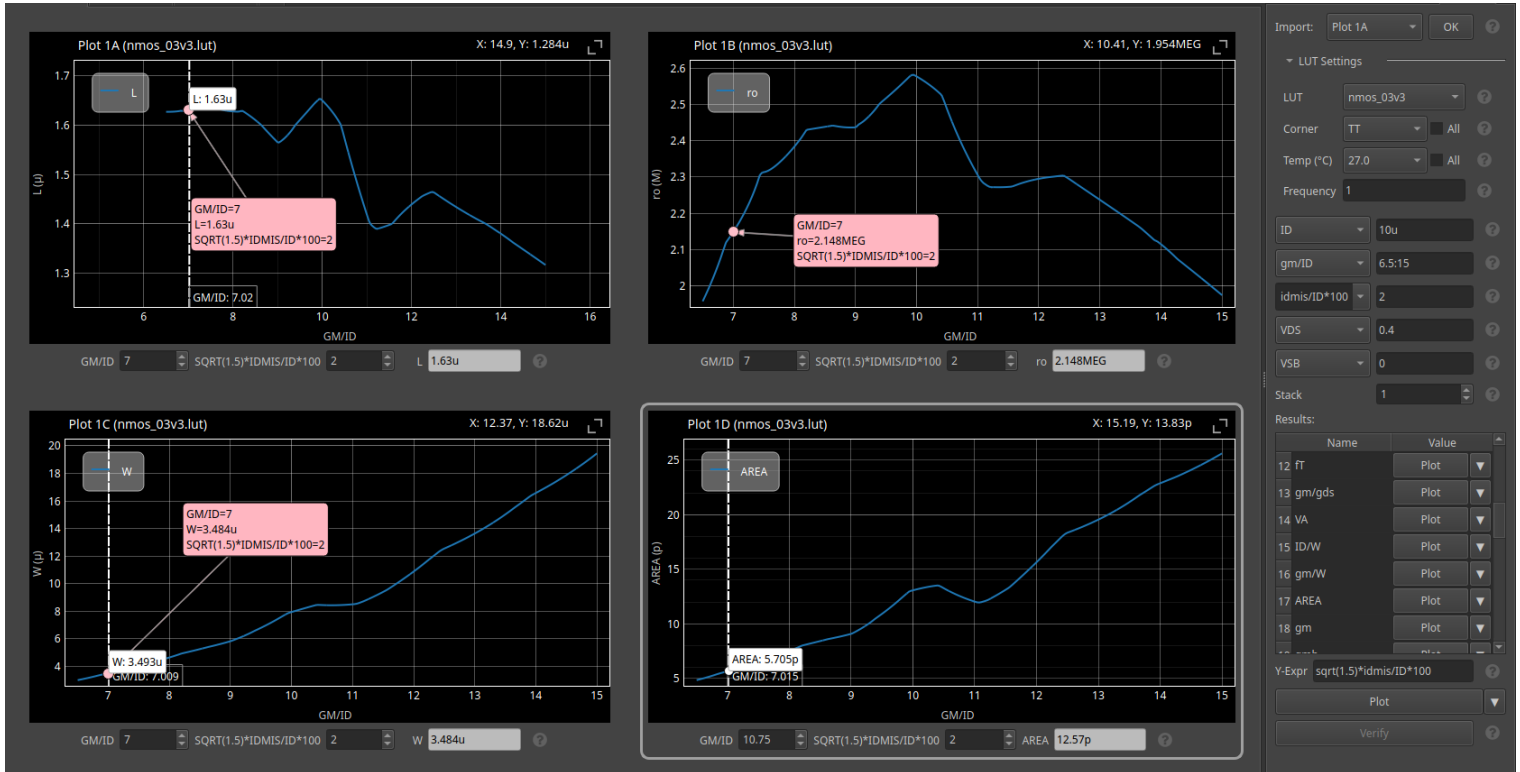
Current mirror sizing

Its V_{ds} will be around the 0.4v at lowest input voltage and

And we want the mismatch to be reasonable around 2%

And we want is r_o to be around 650k to achieve the required CMRR assuming
active load $gm/id = 8$

So we will sweep the area with gm/Id and the most optimum area will be around
 $gm/id = 7$

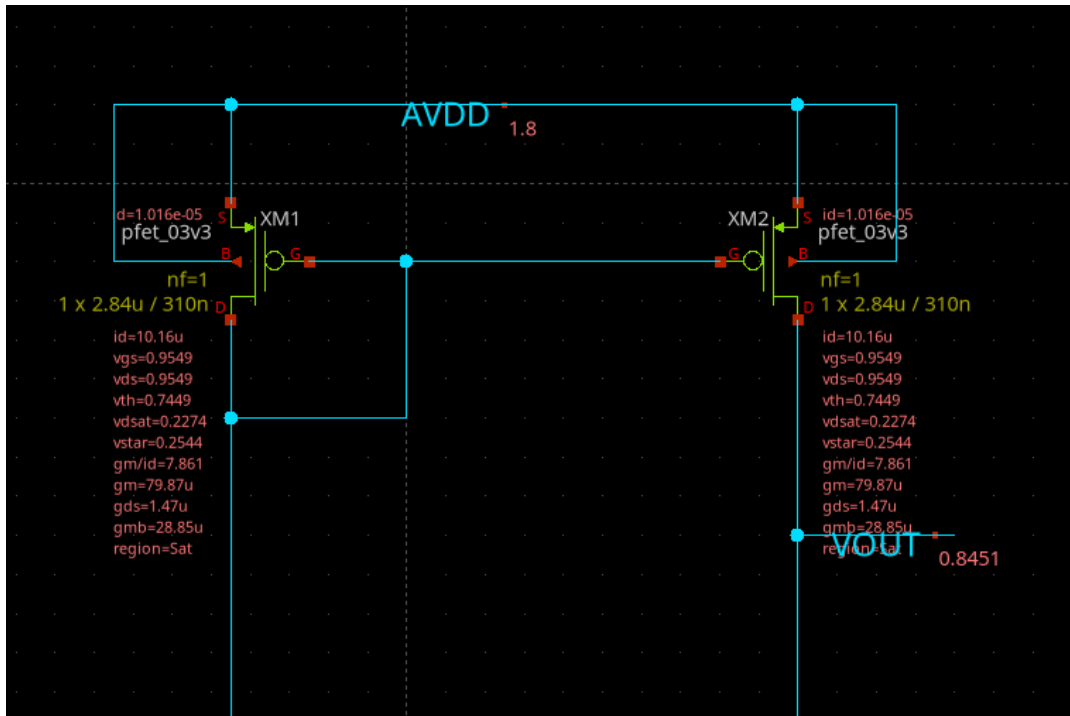


Active load sizing

As we said earlier the gm/id for active load are lower than input mosfets to achieve smaller area



We want to achieve a $r_o = 680k$



We will find that an optimal gm/id is 8 so we can achieve
The CMRR we want and a small area

Plot 2A Plot 2B Plot 2C Plot 2D

Import: Plot 1A OK ?

LUT Settings

LUT pmos_03v3 ?

Corner TT ? All ?

Temp (°C) 27.0 ? All ?

Frequency 1 ?

ID 10u ?

gm/ID 8 ?

ro 689k ?

VDS VGS ?

VSB 0 ?

Stack 1 ?

Results:

	Name	TT-27.0
1	ID	10u
2	IG	N/A
3	L	310n
4	W	2.84u
5	VDB	952.1m
6	VGB	952.1m
7	VGS	952.1m
8	VDS	952.1m

Y-Expr gm/ID*ft ?

Plot ?

Verify ?



Check if requirements are satisfied

```
No. of Data Rows : 101
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

No. of Data Rows : 101
av_diff_db      = 3.419432e+01
av_diff_mag     = 5.125260e+01
bw_diff         = 9.762105e+04
ugf             = 5.000199e+06
p_ugf           = -8.913020e+01
av_cm_db        = -4.975887e+01
av_cm_mag       = 3.251297e-03
bw_cm           = 9.765178e+04
=== AC Results ===
Av Diff: 34.1943 dB, BW: 97621.1 Hz
UGF: 5.0002E+06 Hz
GBW: 5.00333E+06 Hz
PM: 90.8698 deg
Av CM: -49.7589 dB, BW: 97651.8 Hz
CMRR: 83.9532 dB
binary raw file "5t_ota_tb_ol.raw"
binary raw file "/home/ubuntu/lab7/simulation/Lab07_tb.sch/Lab07_tb.raw"
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

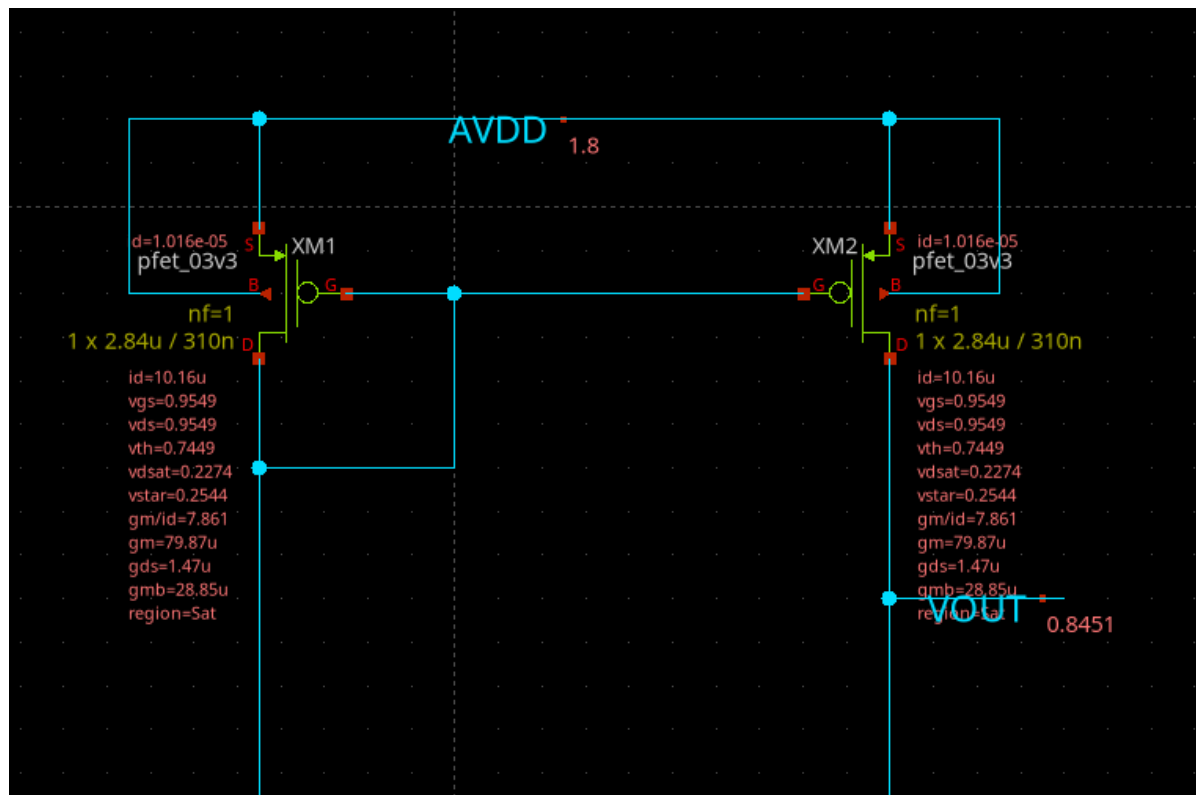
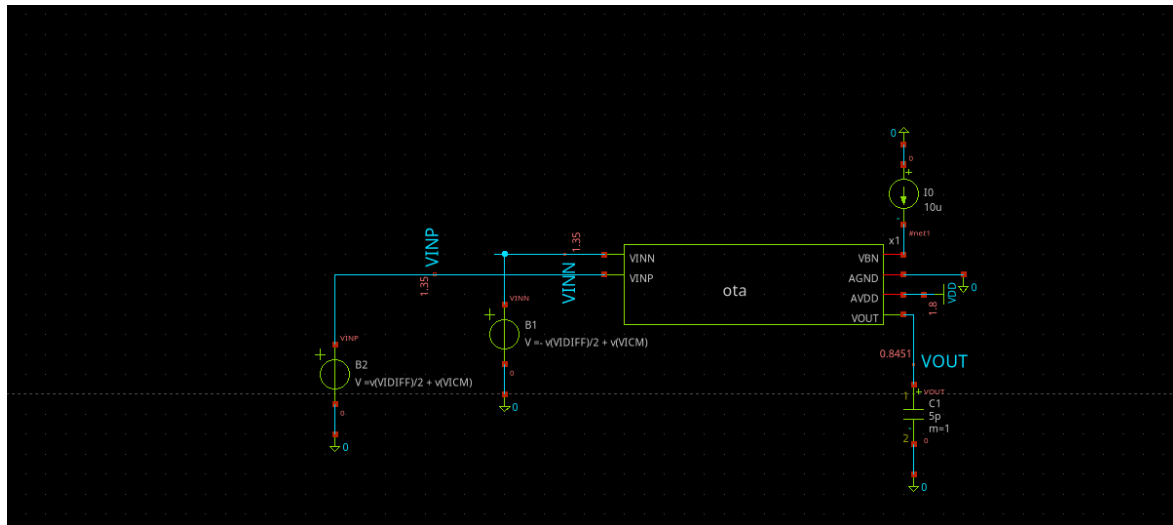
Using SPARSE 1.3 as Direct Linear Solver
OP information in rawfile.

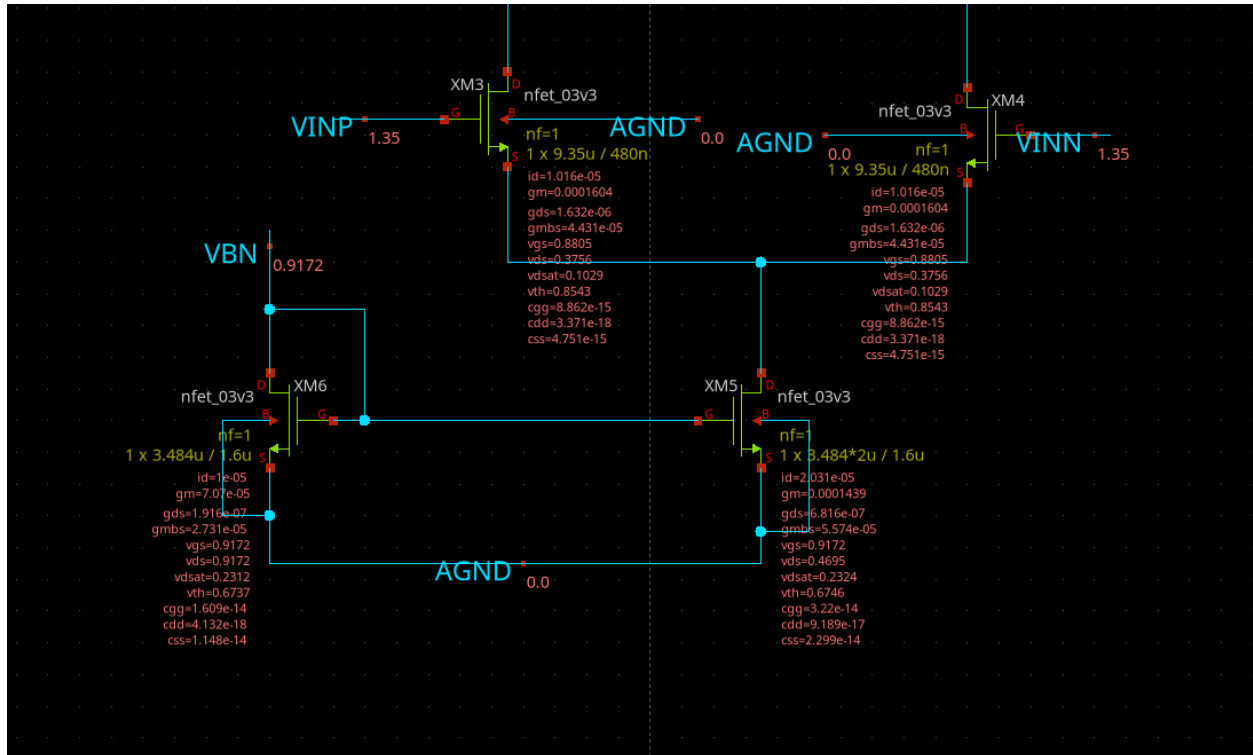
Total analysis time (seconds) = 0.00463223
Total elapsed time (seconds) = 0.005
```



Part 2: Open-Loop OTA Simulation

Schematic

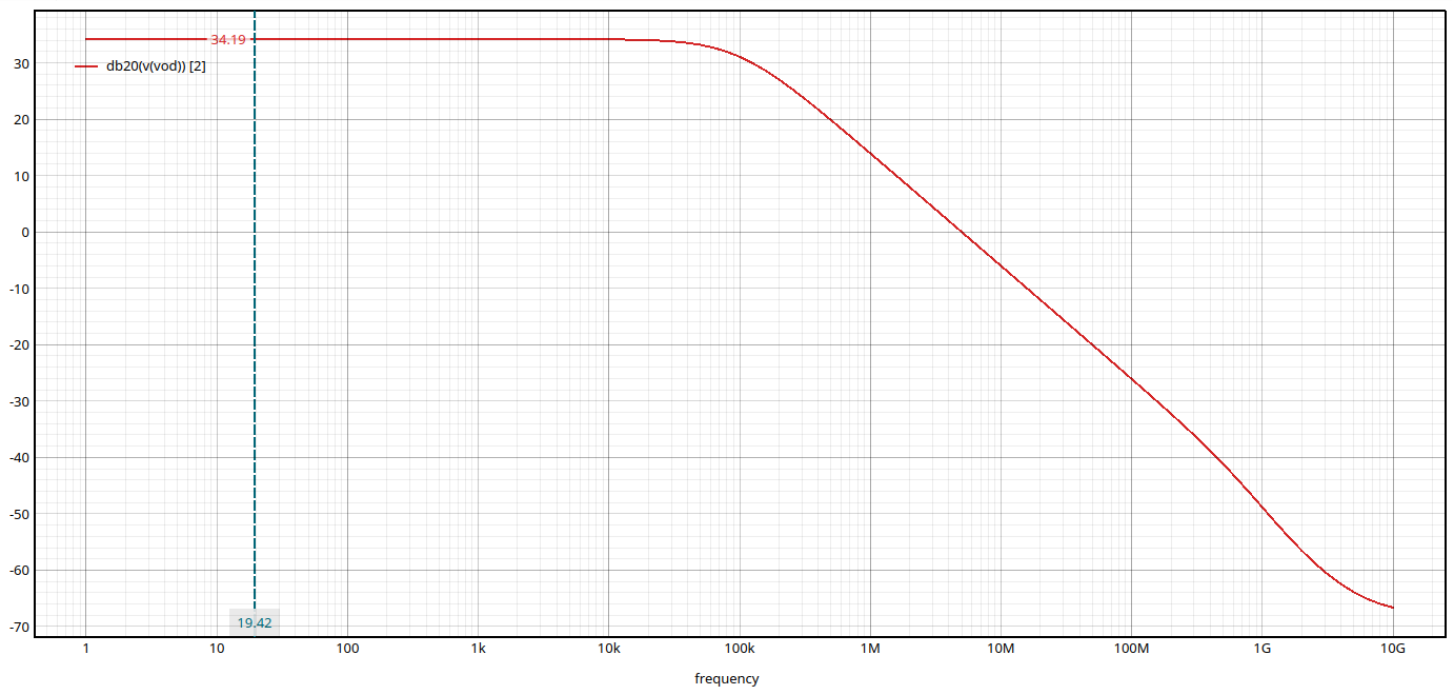




- Yes the currents and gms are exactly the same due to vout following Vf
- The DC level at vout is the same as VF = vgs of the load pmos = 0.845v due to the internal feedback coming from the diode connected mosfet .

Diff small signal ccs

bode plot





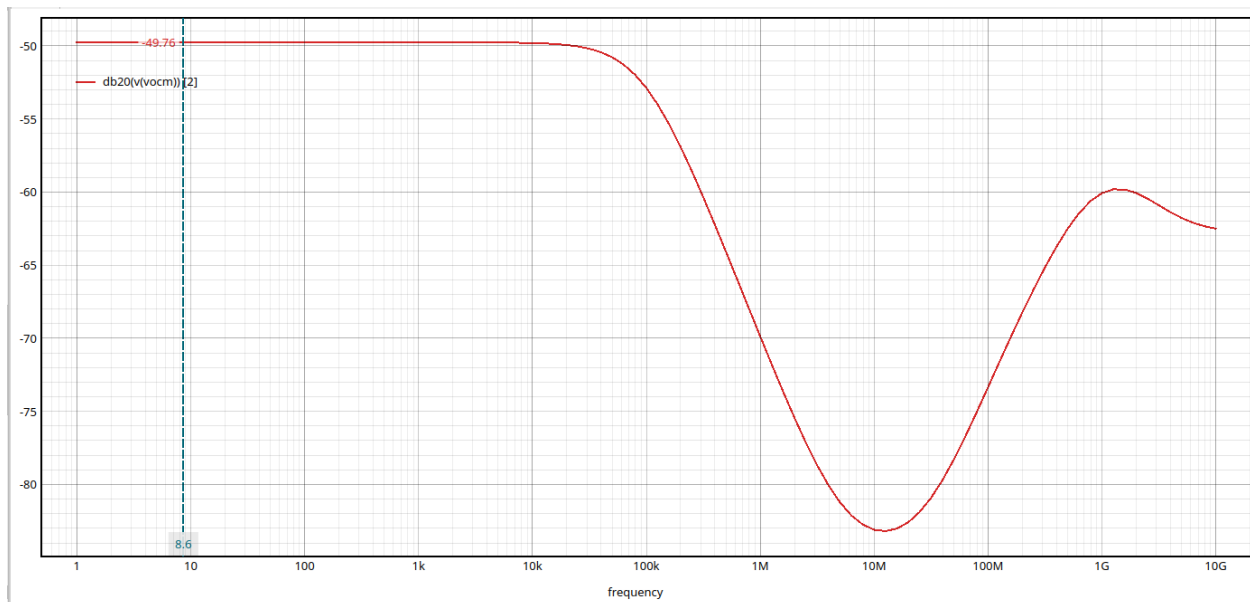
Comparison

hand analysis

$$A_{vd} = g_{m_{1,2}} * (r_{o2} || r_{o4}) = 160\mu s * (600k // 693k) = 51.45 = 34.22db$$

Hand analysis	simulation
34.22db	34.19db

CM small signal ccs



Comparison

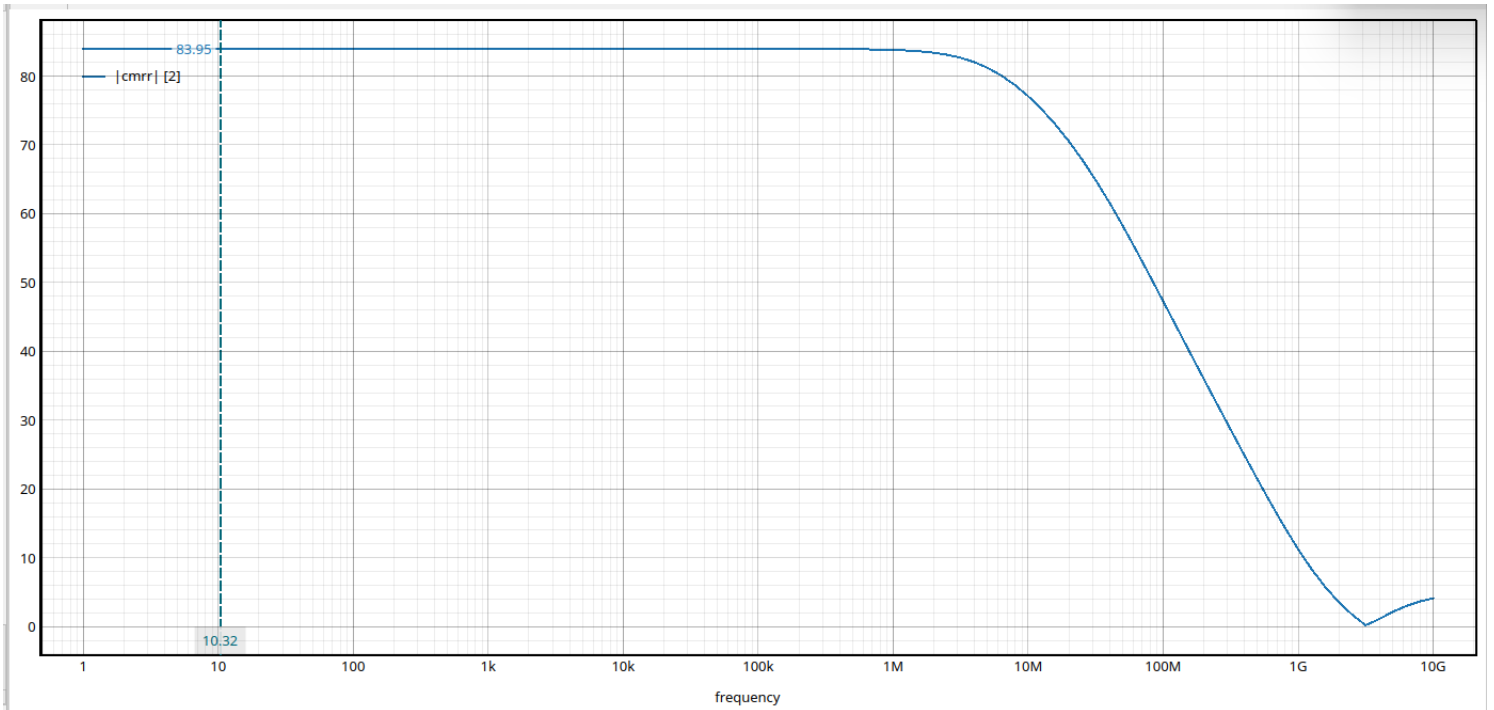
hand analysis

$$A_{vcm} = \frac{1}{g_{m_{3,4}} * R_{ss}} = 80\mu s * 1.4M = 0.008 = -41db$$

Hand analysis	simulation
-41db	-50db



CMRR



Comparison

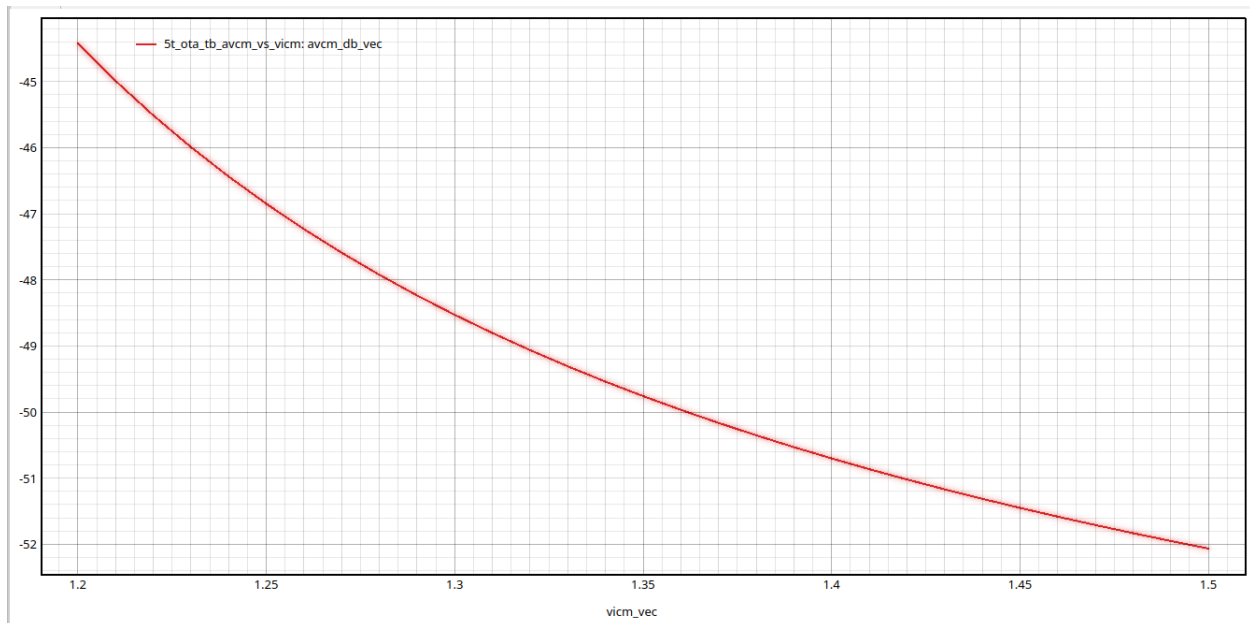
hand analysis

$$\text{CMRR}_{\text{dB}} = 20 \log_{10} \left| \frac{A_{vd}}{A_{vcm}} \right| = 34.22\text{db} + 41\text{db} = 75\text{db}$$

Hand analysis	simulation
75db	83.95db



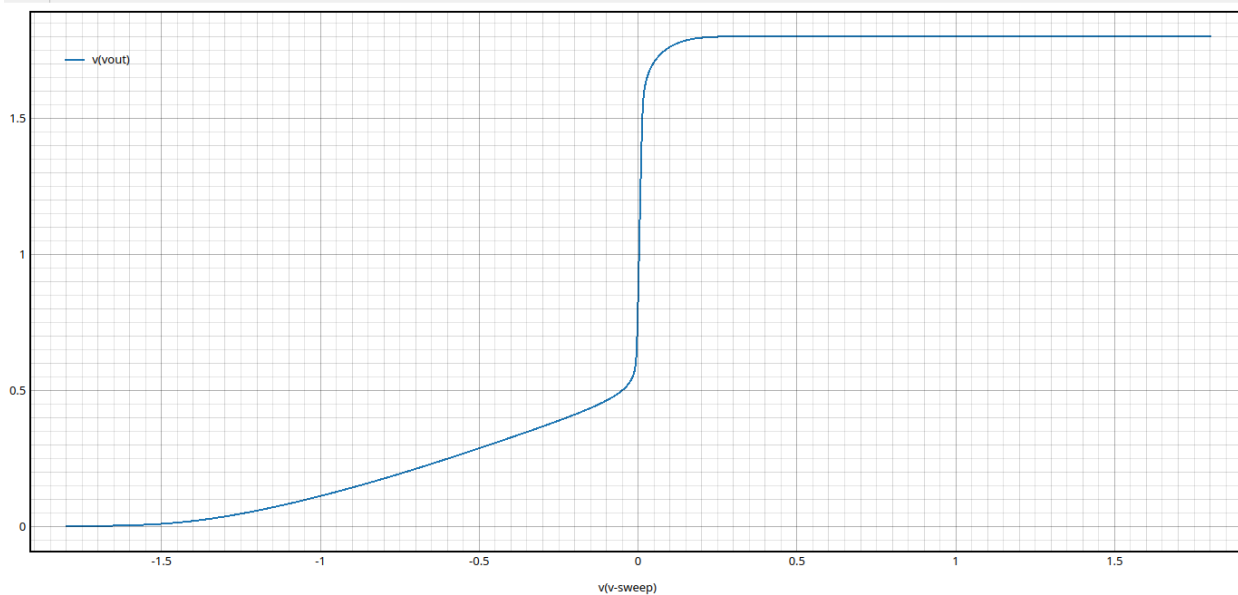
Avcm vs VICM



Decreases as the v_{ds} of the sink nmos increases which leads to increased RSS thus it decreases the A_{vcm}

Diff large signal ccs

Transfer characteristics



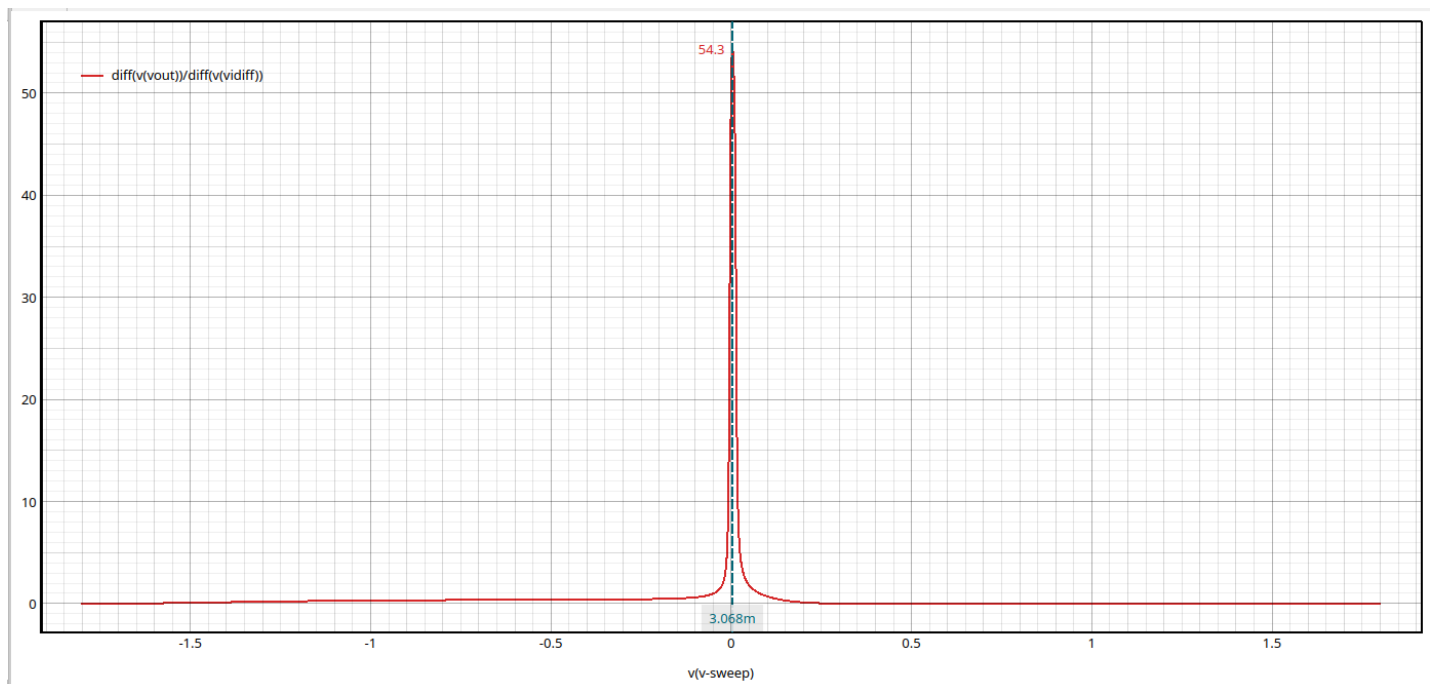


At zero V_{diff} v_{out} is equal to 0.845v which is equal to the v_{gs} of the load mosfet

As due to the internal feedback and the currents in the two branches being equals

So the load mosfet in the v_{out} branch act as a diode connected

Derivative

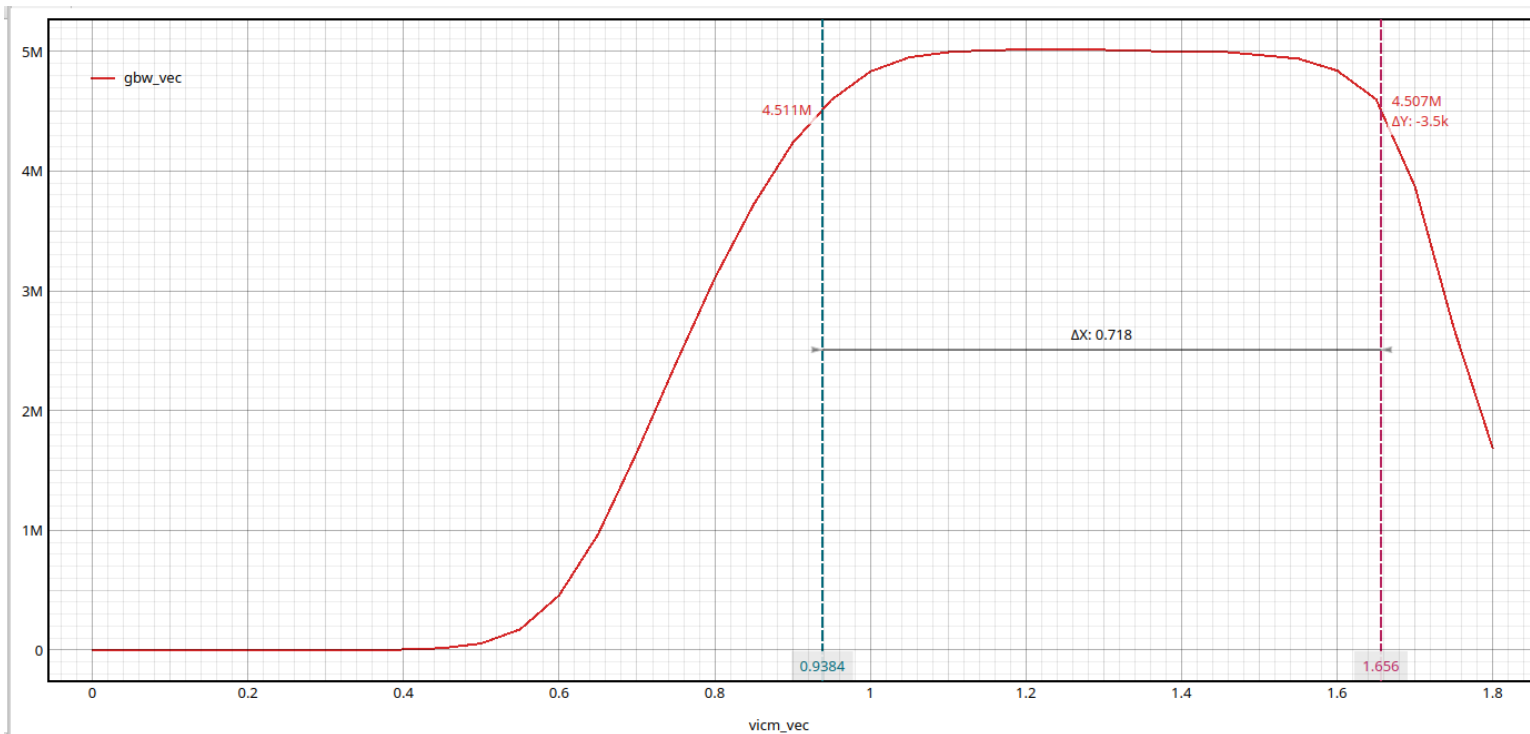


The slope of the transfer characteristic at the middle is nearly equal to the open loop gain

From graph	Simulation results
54.3	51.22



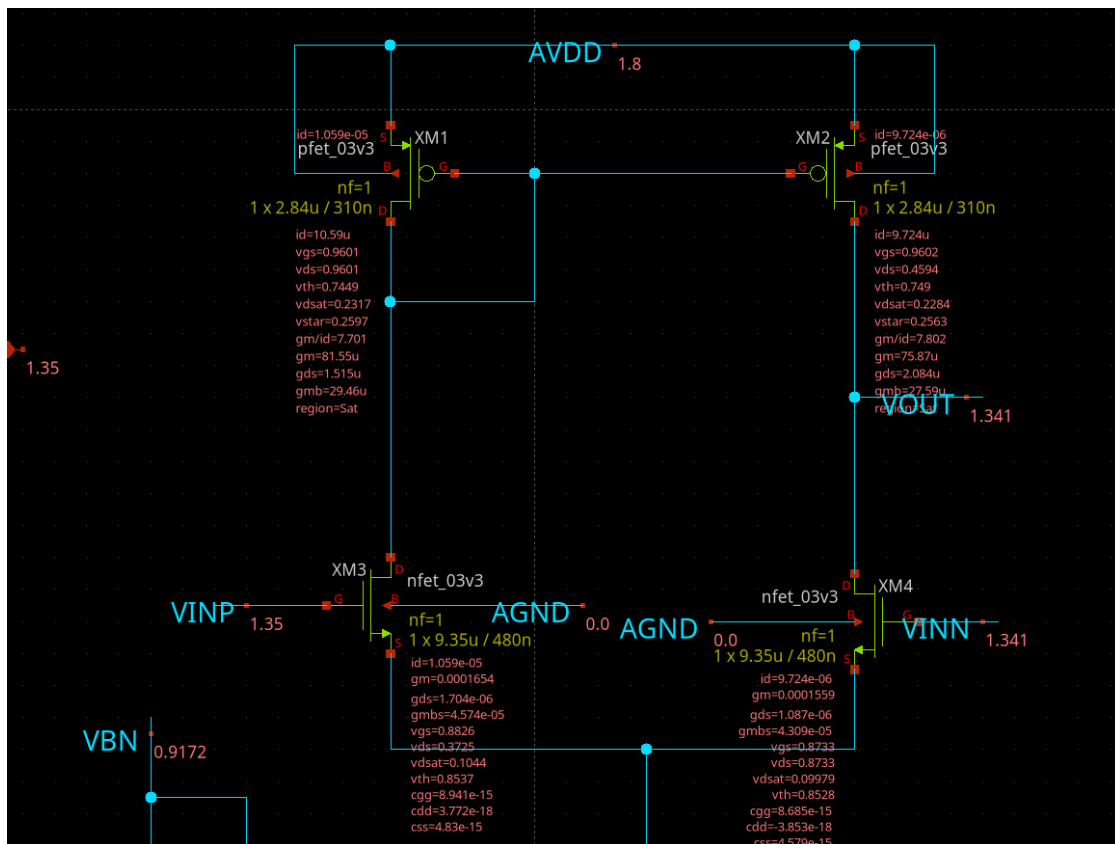
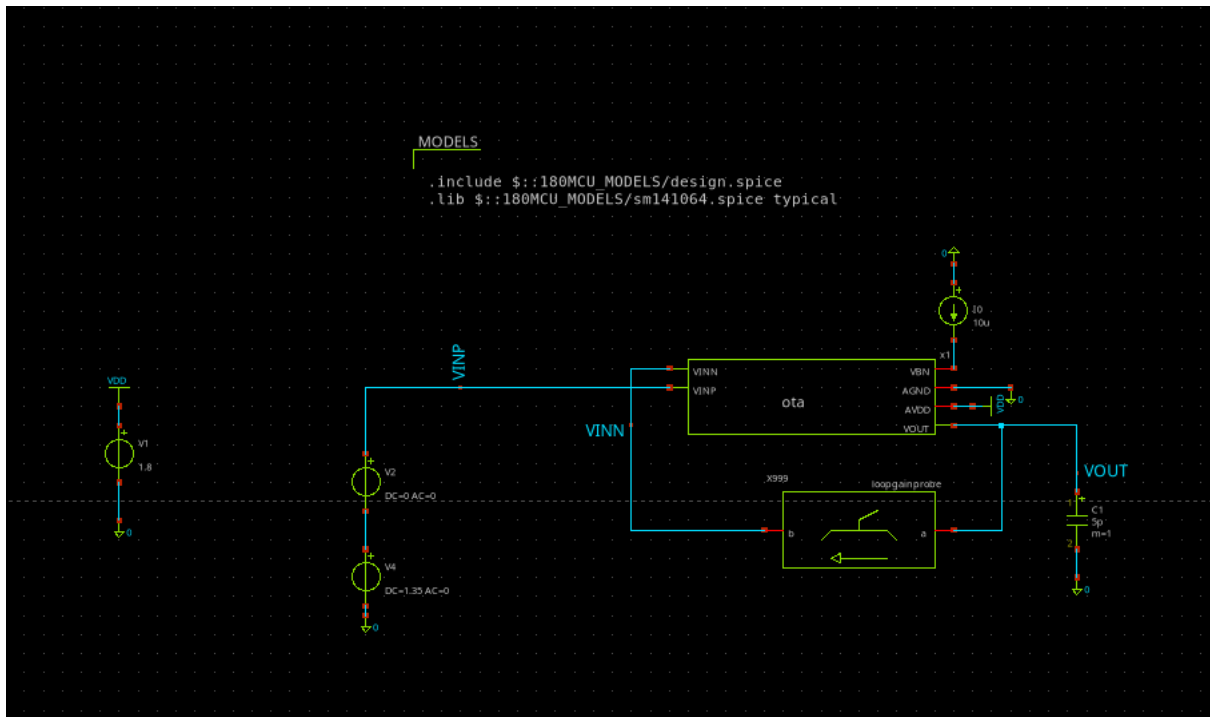
CM large signal ccs (GBW vs VICM)

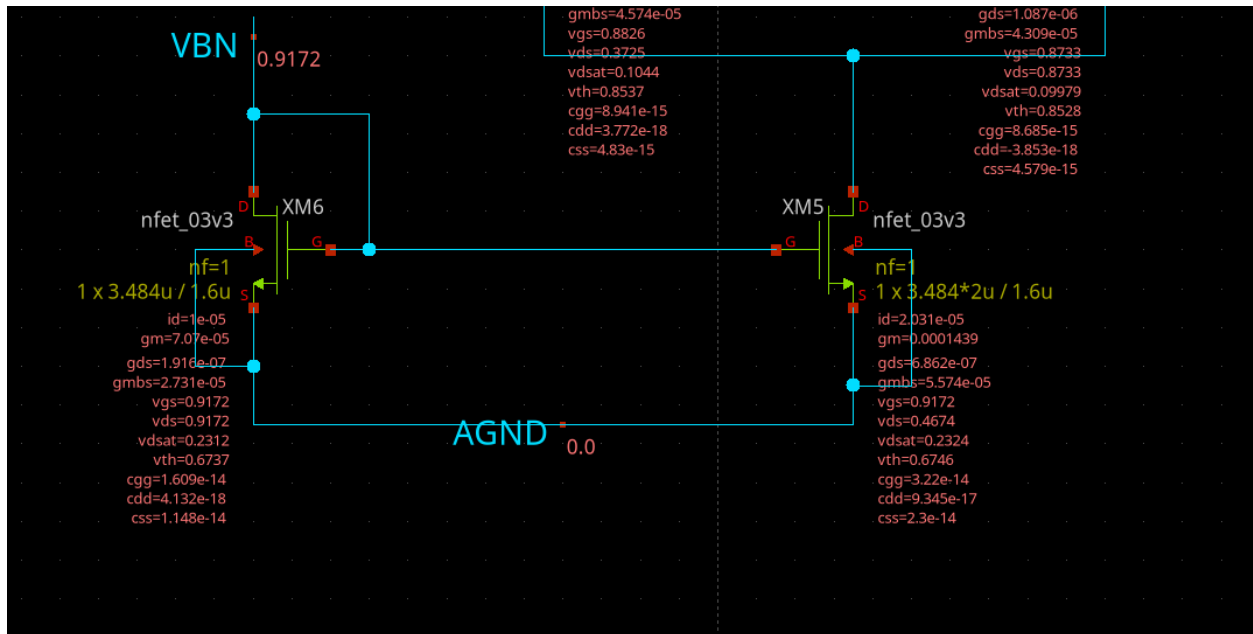




Part 3: Closed-Loop OTA Simulation

Schematic





No the currents aren't exactly equal as vout follows the VINN and VINN follows VINP so the voltage at vout becomes almost equal Vicm DC so the VDS of the PMOS decreases which decreases the current in this branch

Miss match in ID =10.59-9.724=0.866uA

$0.866/10 * 100 = 8.66\%$

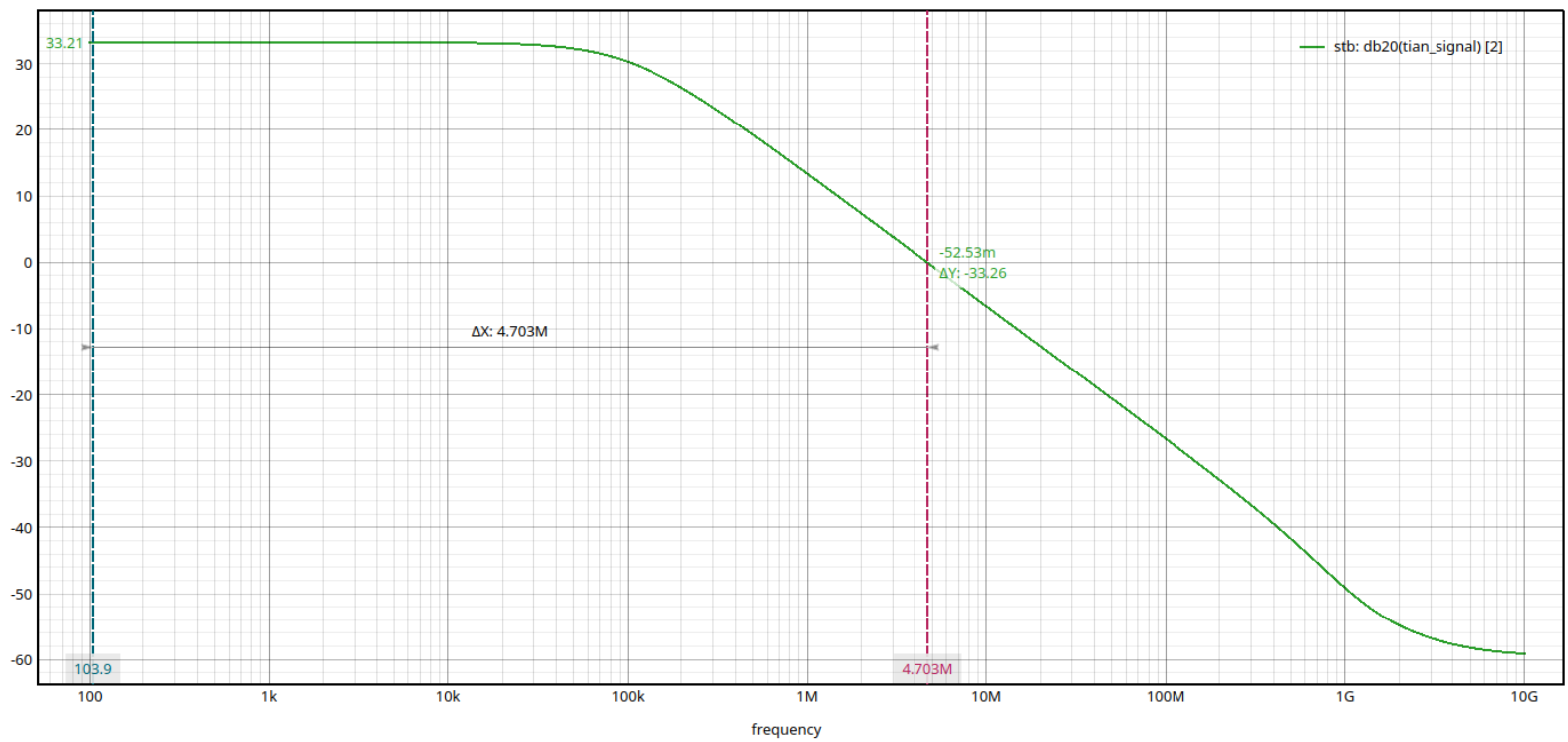
Miss match in gm =81.55-75.87=5.68us

$5.68/80 * 100 = 7.1\%$

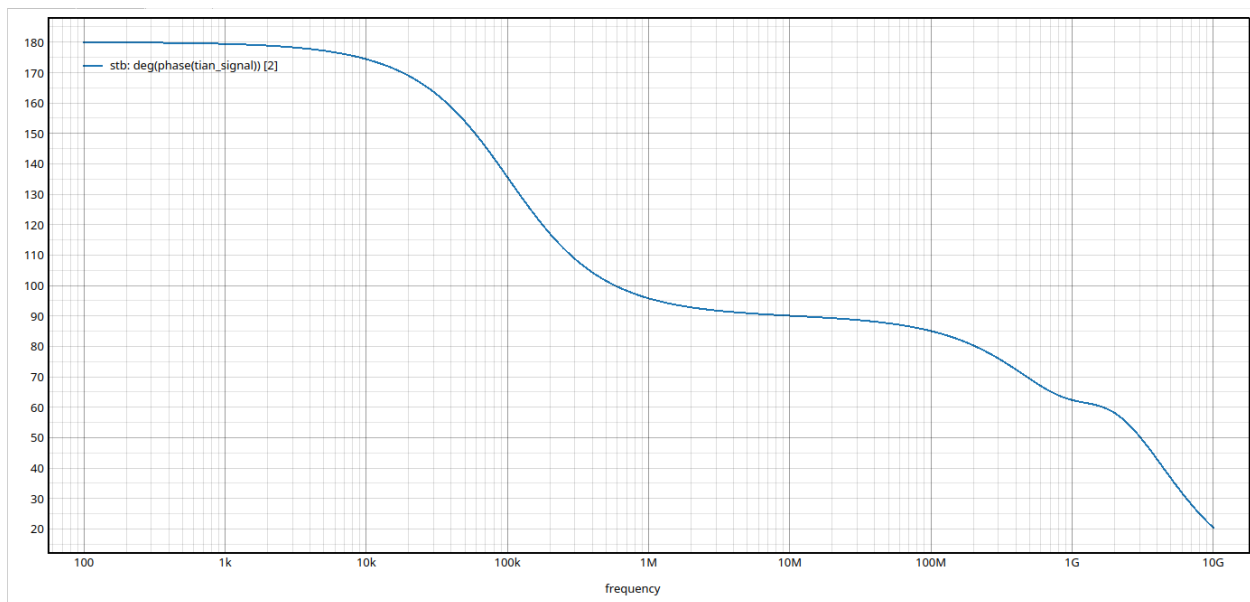


Loop gain

Bode plot



Phase





Parameter	STB loop-gain simulation	Open-loop simulation
DC gain	33.21 dB	34.1943 dB
Gain-crossover frequency (GX) / UGF	4.6744 MHz	5.0002 MHz

Due the feedback and v_{out} following the VINP so the gain decreased a bit due to the subtle change in the biasing point so the this lead to a decrease in UGF

Hand analysis

$$UGF = \frac{gm_{1,2}}{2\pi CL}$$

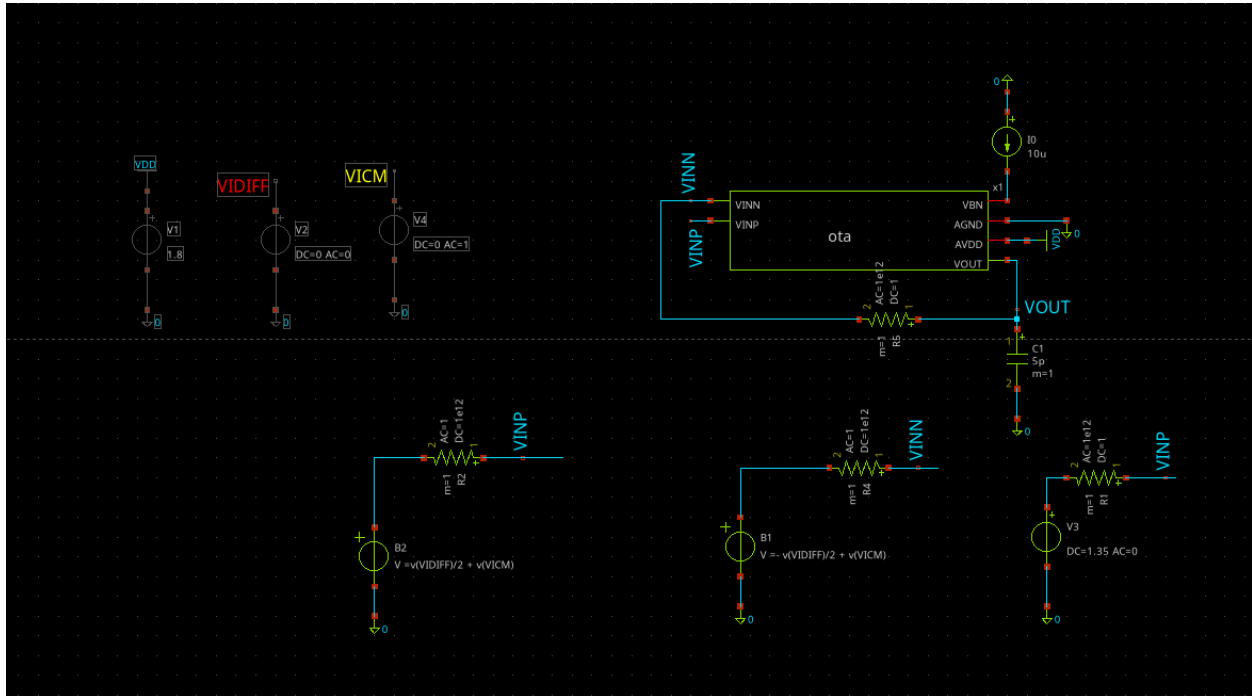
$$UGF = \frac{160\mu s}{2\pi 5pf} = 5.09\text{MHz}$$

Parameter	STB loop-gain simulation	Open-loop simulation	Hand Analysis
DC gain	33.21 dB	34.1943 dB	34.22 dB
Gain-crossover frequency (GX) / UGF	4.6744 MHz	5.0002 MHz	5.09 MHz

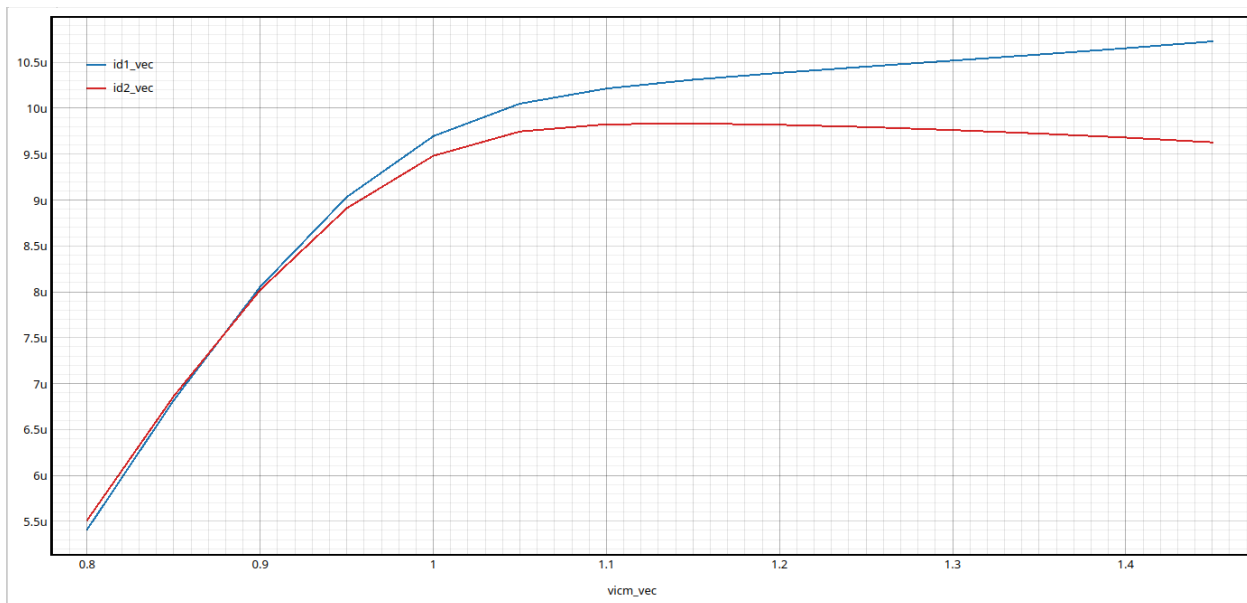


Part 4: Effect of Mismatch on CMRR

Schematic



Plot ID of the input pair overlaid vs VICM

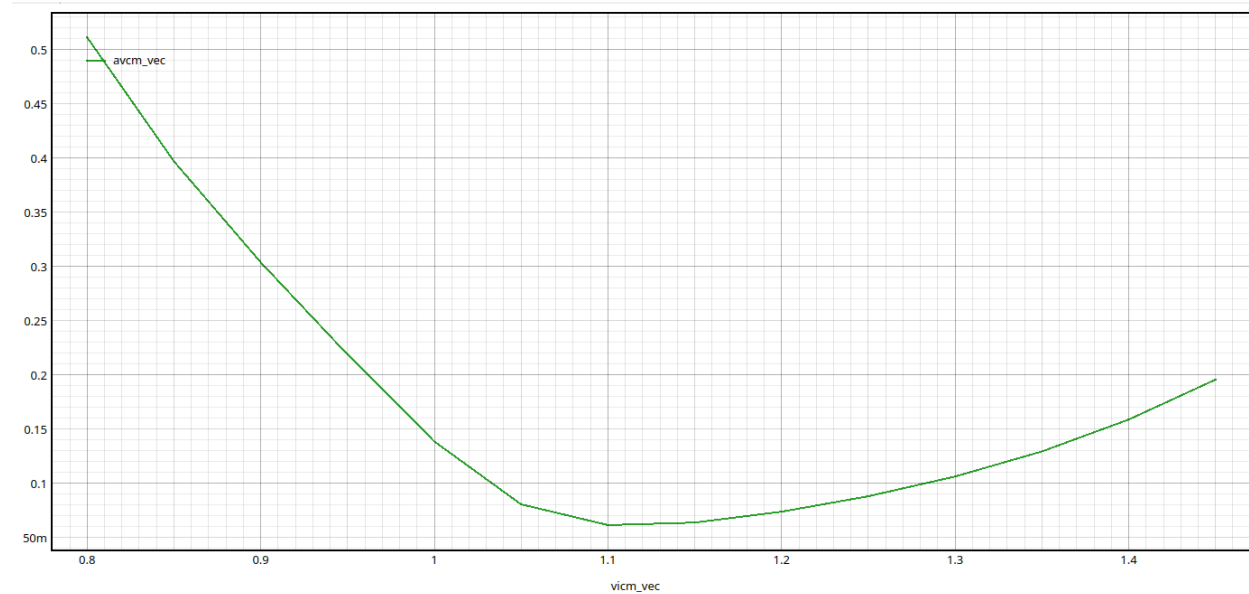




Comment

Due to the increase in V_{cm} and v_{out} following it in DC due to feedback so in this branch the V_{ds} of the PMOS decrease thus the current in this branch decrease while the other increase as they are tied together with I_{SS}

Plot CM gain at 1Hz vs V_{IN}



Comment

First as the V_{cm} increase the node of the source of the input transistor also increases as the current is almost the same so the V_{DS} of I_{SS} increases thus R_{SS} also increases which causes the a_{vcm} to decrease then due the mismatch effect increases as v_{icm} increases Δg_m causing a_{vcm} to increase.

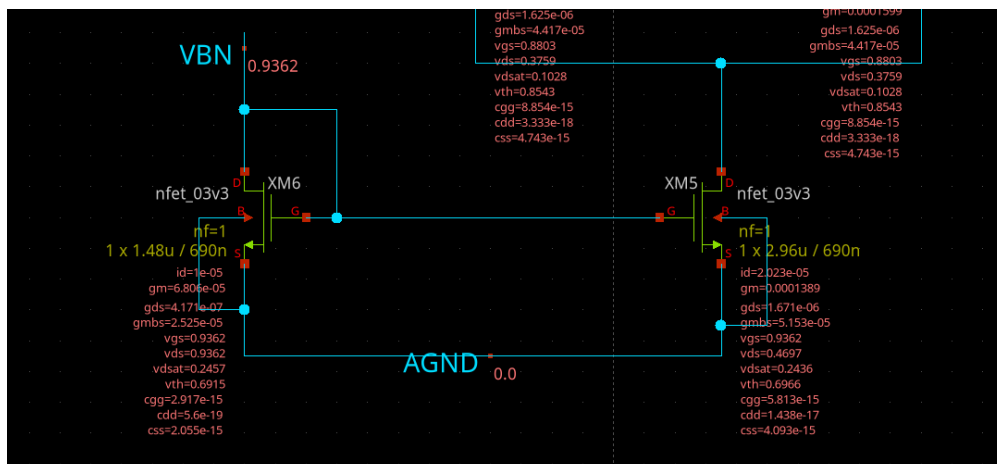
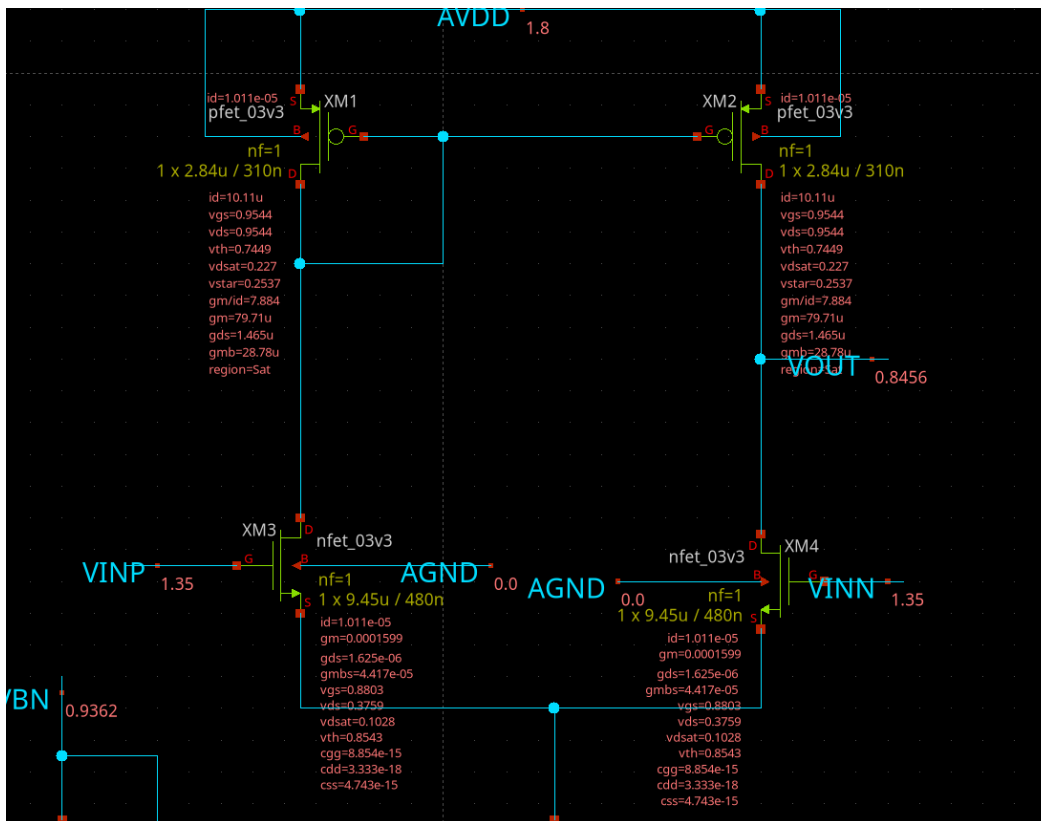


Bonus

By unconstraint the mismatch of the CM we can reduce the area significantly

Here is the new CM sizing

I also needed to increase the W of the input mosfet by 10nm to achieve the GBW



Plot 4A Plot 4B Plot 4C Plot 4D

Import: Plot 1A OK ?

LUT Settings

LUT: nmos_03v3 ?

Corner: TT ? All ?

Temp (°C): 27.0 ? All ?

Frequency: 1 ?

ID: 20u ?

gm/ID: 7 ?

ro: 625k ?

VDS: 0.47 ?

VSB: 0 ?

Stack: 1 ?

Results:

	Name	TT-27.0
1	ID	20u
2	IG	N/A
3	L	690n
4	W	2.96u
5	VDB	470m
6	VGB	933.8m
7	VGS	933.8m
8	VDS	470m

Y-Expr: $\sqrt{1.5} * idmis / ID * 100$?

Plot ?

Verify ?



```
View data
Using SPARSE 1.3 as Direct Linear Solver
No. of Data Rows : 101
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver
No. of Data Rows : 101
av_diff_db      = 3.421362e+01
av_diff_mag     = 5.136662e+01
bw_diff         = 9.737054e+04
ugf             = 4.998409e+06
p_ugf           = -8.913440e+01
av_cm_db        = -4.199183e+01
av_cm_mag       = 7.950753e-03
bw_cm           = 9.737429e+04
=== AC Results ===
Av Diff: 34.2136 dB, BW: 97370.5 Hz
UGF: 4.99841E+06 Hz
GBW: 5.0016E+06 Hz
PM: 90.8656 deg
Av CM: -41.9918 dB, BW: 97374.3 Hz
CMRR: 76.2054 dB
binary raw file "5t_ota_tb_ol.raw"
binary raw file "/home/ubuntu/lab7bonus/simulation/Lab07_tb.sch/Lab07_tb.raw"
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver
OP information in rawfile.
```

Dismiss Save As